

Code : 303205

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BCA 2nd Semester Exam., 2014

OPERATING SYSTEM AND UNIX

Time : 3 hours akubihar.com Full Marks : 60

Instructions: akubihar.com

- (i) All questions carry equal marks.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Choose the correct answer/Fill in the blank (any six) :

- (a) UNIX is a
 - (i) single-user and CUI-based operating system
 - (ii) multiuser and GUI-based operating system
 - (iii) multiuser and single-tasking operating system
 - ☒ (iv) multiuser and multitasking operating system

- (b) wc command
 - (i) is used for counting the character of a word akubihar.com
 - (ii) counts the line of a file
 - (iii) counts the files of a directory
 - (iv) is a wrong command

- (c) To erase the entire word in vi editor, we use
 - (i) Ctrl-w
 - (ii) Ctrl-c
 - (iii) Alt-w
 - (iv) Ctrl-d

- (d) Which was the first UNIX shell?
 - (i) Bash shell
 - (ii) Korn shell
 - ☒ (iii) Bourne shell
 - (iv) X-shell

- (e) To remove a directory — command is used in UNIX.

- (f) The octal number used with chmod command to make a file readable, writable and executable to the owner is

- (i) 000
- (ii) 722
- (iii) 755
- (iv) 777

- (g) The process is **akubihar.com**
- (i) job in execution
 - (ii) job in RAM
 - (iii) job in waiting
 - (iv) None of the above
- (h) How many types of scheduler are there?
- (i) 2
 - (ii) 3
 - (iii) 4
 - (iv) 5
- (i) The SJF stands for
- (i) shortest job first
 - (ii) simple job first
 - (iii) system job first
 - (iv) second job first
- (j) To see the contents of a directory, we use the command
- (i) ls
 - (ii) lp
 - (iii) dir
 - (iv) list

2. Answer any *three* short answer-type questions :

- (a) What is the role of operating system?
- (b) Differentiate between page and segment.
- (c) Describe about process.

- (d) Deals about the three modes of vi editor. How would you switch from one mode to another? **akubihar.com**
- (e) Write the syntax of the following commands :
- (i) cmp
 - (ii) cat
 - (iii) rmdir
 - (iv) comm
3. How many types of scheduling algorithm are there? Describe them in detail.
4. Describe the paging techniques. What is the need of page table?
5. Compare UNIX features with Windows. Give your views which is better.
6. How many types of shell are used in UNIX? Describe them. **akubihar.com**
7. Write short notes on any *three* of the following :
- (a) Fragmentation
 - (b) GUI **akubihar.com**
 - (c) UNIX file system
 - (d) Shell
 - (e) Multithreading

BCA 2nd Semester Exam., 2016

OPERATING SYSTEM AND UNIX

Time : 3 hours

Full Marks : 60

Instructions :

- (i) All questions carry equal marks.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

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1. Answer the following/Fill in the blank (any six) :

- (a) How do you find, whether your system is 32-bit or 64-bit?
- (b) What does preemptive mean?
- (c) What three requirements must a solution, to the critical-section problem, satisfy?
- (d) What is meant by 'automatic job sequencing'?
- (e) _____ is a technique of temporarily removing inactive programs from the memory of computer system. akubihar.com
- (f) What resources must be protected by the operating system?

- (g) How do you set environment variable which will be accessible from sub shell?
- (h) What is the command used to replace many characters in Vi Editor?
- (i) What is meant by contiguous allocation?
- (j) Where should the disk directories be placed?

2. Answer any three short-answer type questions :

- (a) When do page faults occur? Describe the actions taken by the operating system when a page fault occurs. akubihar.com
- (b) Why are segmentation and paging sometimes combined into one scheme?
- (c) Explain the File Access Modes in UNIX.
- (d) What are the four necessary conditions of deadlock prevention?
- (e) What advantage is there in having different time-quantum sizes on different levels of a multilevel queueing system?

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3. Write down the advantages and disadvantages of LINUX. Also discuss the components of LINUX.

4. Explain any three policies for process scheduling that uses resource consumption information. What is response ratio?
5. Discuss memory protection and CPU protection techniques.
6. What is shell? How many types of shells are used in UNIX? Describe.
7. Write short notes on any *three* of the following :
 - (a) Access control list
 - (b) Process scheduling in LINUX
 - (c) Role of UNIX system administrator
 - (d) Semaphore
 - (e) Reentrancy

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BCA 2nd Semester Theory Examination, 2017

Operating System and Unix

Time : 3 hours

Full Marks : 60

Instructions :

- (i) *There are seven questions in this Paper. All questions carry equal marks.*
 - (ii) *Attempt five questions in all.*
 - (iii) *Question No. 1 & 2 is compulsory.*
1. (a) How do you find whether your system is 32 bit or 64 bit?
- (b) What does pre-emptive means?
- (c) What three requirements must a solution to the critical-section problem satisfy?
- (d) What is "automatic job sequencing" means?
- (e) is a technique of temporarily removing inactive programs from the memory of computer system.
- (f) What resources must be protected by the operating system?
- (g) How do you set environment variable which will be accessible from sub shell?
- (h) What is the command used to replace many characters in Vi Editor?

P.T.O.

(i) What is contiguous allocation means?

(j) Where should the disk directories be placed?

2. Answer any three short type of answer type questions:

- (a) When do page faults occur? Describe the actions taken by the operating system when a page fault occurs?
- (b) Why are segmentation and paging sometimes combined into one scheme?
- (c) Explain the File Access Modes in Unix?
- (d) What are the four necessary conditions of deadlock prevention?
- (e) What advantage is there in having different time-quantum sizes on different levels of a multilevel queuing system?

3. Write down the advantages and disadvantages of Linux? Also discuss the components of Linux?

4. Explain any three policies for process scheduling that use resource consumption information. What is response ratio?

5. Discuss Memory protection and CPU protection techniques.

6. What is Shell? Describe how many types of shell are used in Unix?

Code : 303205

2

7. Write a Short Notes on any three of the following:

- (a) Access Control List
- (b) Process scheduling in Linux
- (c) Role of Unix System administrator
- (d) Semaphore
- (e) Reentrancy

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BCA 2nd Semester Exam., 2018

OPERATING SYSTEM AND UNIX

Time : 3 hours

Full Marks : 60

Instructions :

- (i) All questions carry equal marks.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six questions as directed :

- (a) How do you find whether your system is 32-bit or 64-bit?
 - (b) Page stealing
 - (i) is a sign of an efficient system
 - (ii) is taking page frames from other working sets
 - (iii) should be the tuning goal
 - (iv) is taking larger disk spaces for pages paged out
 - (v) None of the above
- (Choose the correct option)

(Turn Over)

(2)

- (c) The main reason to encrypt a file is to _____.
(Fill in the blank)
 - (d) Before proceeding with its execution, each process must acquire all the resources : needs is called _____.
(Fill in the blank)
 - (e) _____ is a technique of temporarily removing inactive programs from the memory of computer system.
(Fill in the blank)
 - (f) What is race condition?
 - (g) How do you set environment variable which will be accessible from subshell?
 - (h) What is the command used to replace many characters in vi editor?
 - (i) What does SSTF stand for?
 - (j) The memory allocation scheme subject to 'external' fragmentation is
 - (i) segmentation
 - (ii) swapping
 - (iii) pure demand paging
 - (iv) multiple fixed contiguous partition
- (Choose the correct option)

(Continue)

2. Answer any three questions in short :

- (a) Give the difference between Multi-programming and Multiprocessing.
- (b) Differentiate between Preemptive and Non-preemptive scheduling.
- (c) Explain the file access modes in UNIX.
- (d) Describe three methods for passing parameters needed by system calls.
- (e) Describe the differences among short-term, medium-term and long-term scheduling.

3. List the steps needed to perform page replacement. Explain the different page replacement policies. Also list out the main requirements which should be satisfied by a page replacement policy.

4. What is meant by inter-process communication? Explain the two fundamental models of inter-process communication.

5. The Linux kernel does not allow paging out of kernel memory. What effect does this restriction have on the kernel's design? What are the two advantages and disadvantages of this design decision?

(Turn Over)

6. What is shell? Describe how many types of shell are used in UNIX.

7. Write short notes on any three of the following :

- (a) Semaphore
- (b) Belady's anomaly
- (c) Translation Lookaside Buffer (TLB)
- (d) Different modes in vi editor
- (e) Garbage collection

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(2)

BCA 2nd Semester Exam., 2019

OPERATING SYSTEM AND UNIX

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six of the following : $2 \times 6 = 12$

- (a) What is the need of Operating System?
- (b) What is a process?
- (c) Discuss the concept of thrashing.
- (d) What is a thread?
- (e) Define critical section.
- (f) How is the time set and shown in Linux?

(Turn Over)

AK9/776

- (g) List different components of a Linux system.
- (h) Why is file protection necessary?
- (i) Define the physical address space.
- (j) Define anti-virus and its types.

2. Answer any three of the following : $4 \times 3 = 12$

- (a) What is an operating system? What are the various services provided by the operating systems?
- (b) What are the various deadlock conditions? Explain in detail. Also, write banker algorithm.
- (c) Discuss in detail internal and external fragmentation.
- (d) Explain basic features of UNIX/Linux.
- (e) Differentiate between the following :
 - (i) Paging and demand paging
 - (ii) LRU and optimal page replacement algorithm

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(Continue)

(3)

3. Explain the different states of the process.
What is the purpose of process control block? 12
4. What is a file? What are different types of
file? Explain various file access methods. 12
5. What are semaphores? How are they
implemented? 12
6. Write a procedure to log in and log out for
Linux operating system. Write steps for
changing shell prompt. 12
7. Discuss the structure of LINUX Operating
System. 12

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BCA 2nd Semester Exam., 2021

OPERATING SYSTEM AND UNIX

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six of the following : $2 \times 6 = 12$

- (a) What is the relationship between operating systems and computer hardware?
- (b) What is the difference between job and process?
- (c) What is batch processing system?
- (d) Define real time system.
- (e) Define process.
- (f) Define thread.
- (g) What is memory management?
- (h) Define deadlock.

(i) What do you mean by paging?

(j) What is scheduling?

2. Answer any three of the following : $4 \times 3 = 12$

- (a) What is the difference between kernel mode and user mode?
- (b) What criteria affect the scheduler's performance?
- (c) When does a page fault occur?
- (d) What are the advantages and disadvantages of batch processing system?
- (e) Write some features of Linux Operating System. <https://www.akubihar.com>

3. Describe system calls and its type. 12

4. Describe the difference among user level, kernel level and hybrid level thread. 12

5. Discuss the methods for handling deadlock. 12

6. Explain the various address translation techniques in paging. 12

7. Discuss the file management technique of Linux. 12

Code : 303205

(2)

BCA 2nd Semester Exam., 2022

OPERATING SYSTEM AND UNIX

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1** and **2** are compulsory.

1. Answer any six of the following questions :

2×6=12

- (a) What is caching?
- (b) What is scheduler?
- (c) What is multiprogramming?
- (d) What is internal fragmentation?
- (e) What is the use of vi editor?
- (f) What is the use of tsort command in Unix?
- (g) What is vim?

AK23/701

(Turn Over)

(h) What are device drivers?

(i) What is spooling?

(j) What is thrashing?

2. Answer any three of the following questions :

4×3=12

- (a) Describe the difference among long term, medium term and long-term scheduler.
- (b) Describe the difference between internal fragmentation and external fragmentation.
- (c) Describe the difference between safe and unsafe states of deadlock.
- (d) What are the features of Linux?
- (e) What is the meaning of shell and what is its purpose?

Answer any three of the following questions :

12×3=36

- 3. Briefly discuss scheduling algorithms.
- 4. Discuss the classical problem of synchronization.
- 5. Discuss the page replacement algorithms.

AK23/701

(Continued)

(3)

- ✓ 6. Discuss some Linux commands and its utilities.
7. Discuss the role of Unix system administrator.

BCA 2nd Semester Exam., 2023

OPERATING SYSTEM AND UNIX

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1 and 2** are compulsory.

1. Answer any six of the following : $2 \times 6 = 12$

- (a) How to define scheduling?
- (b) What is the use of I/O buffering?
- (c) What is the purpose of threads?
- (d) What is the purpose of precedence graph?
- (e) What is the use of memory management unit?
- (f) What is the difference between frame and page table?

- (g) What is the use of mv and wc command?
- (h) What is the meaning of shell?
- (i) With respect to shell, what is the purpose of job control?
- (j) What is the purpose of pipes and filters?

2. Answer any three of the following : $4 \times 3 = 12$

- (a) Define timesharing differs from multiprogramming. If so, how?
- (b) List two programming examples of multithreading giving improved performance over a single-threaded solution.
- (c) What are the steps required to handle a page fault in demand paging?
- (d) What is the use of vi editor? Write down the features of vi editor.
- (e) With respect to Linux utilities, what is the purpose of permission modes and standard files?

3. What is critical section problem? Discuss the solution to producer-consumer problem using semaphores.

12

4. What is the necessary condition of deadlock? Discuss deadlock detection (Banker's algorithm) with an example. 12

5. With respect to Linux, discuss Linux process and thread management. 12

6. Consider the following reference string :

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0,
3, 2, 1, 2, 0, 1, 7, 0, 1

Assume there are three frames. Apply LRU replacement algorithm to the reference string above and find out how many page faults are produced. Illustrate the LRU page replacement algorithm in detail and also two feasible implementations of the LRU algorithm. 12

7. What is process scheduler? Discuss CPU scheduling algorithms with examples. 12

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BCA 2nd Semester Exam., 2025

OPERATING SYSTEM AND UNIX

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. **1** and **2** are compulsory.*

1. Choose the correct answer (any six) : $2 \times 6 = 12$

(a) The layer between the hardware and the user program is

- (i)* operating environment
- (ii)* system environment
- (iii)* operating system
- (iv)* All of the above
- (v)* None of the above

26AK/201

(Turn Over)

(2)

(b) The resource management can be done in an operating system via

- (i) space division multiplexing
- (ii) time division multiplexing
- (iii) both time and space division multiplexing
- (iv) All of the above
- (v) None of the above

(c) _____ operating system does not implement multitasking.

- (i) Windows XP
- (ii) Windows NT
- (iii) Windows 98
- (iv) MS-DOS
- (v) None of the above

(d) A platform for other software to run on is called

(i) operating system

(ii) system software

(iii) application software

(iv) All of the above

(v) None of the above

(e) The systems that allow only one process execution at a time are

(i) unitasking systems

(ii) uniprogramming systems

(iii) uniprocessing systems

(iv) Both (i) and (iii)

(v) None of the above

(4)

(f) Interprocess communication is

- (i) communication between two processes
- (ii) communication between two threads of the same process
- (iii) communication within the process
- (iv) Both (i) and (iii)
- (v) None of the above

(g) The number of processes completed per unit time is called

- (i) capacity
- (ii) output
- (iii) Both (i) and (ii)
- (iv) throughput
- (v) efficiency

(5)

(h) A process terminates when

- (i) it is removed from all, but the job queue .
 - (ii) its process control block is never deallocated
 - (iii) its process control block is deallocated
 - (iv) it is removed from all queues
 - (v) None of the above
- (i) Which one of the following is a synchronization tool?
- (i) Pipe
 - (ii) Thread
 - (iii) Socket
 - (iv) Semaphore
 - (v) Both (ii) and (iv)

(6)

(j) Multiprogramming of the computer system increases

(i) memory

(ii) storage

(iii) CPU utilization

(iv) cost

(v) All of the above

2. Answer any *three* of the following : $4 \times 3 = 12$

(a) What is CPU scheduling?

(b) What is deadlock problem?

(c) What are logical and physical addresses in operating system?

(d) What are the components of Linux?

(e) What is piping in Linux?

3. Explain system calls.

12

26AK/201

(Continued)

(7)

4. What is a thread? Explain the difference between process and thread. 12
5. What does contiguous memory allocation mean? 12
6. What is interprocess communication? Explain synchronization in interprocess communication. 12
7. What are the basic features of Linux? Explain its components. 12

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